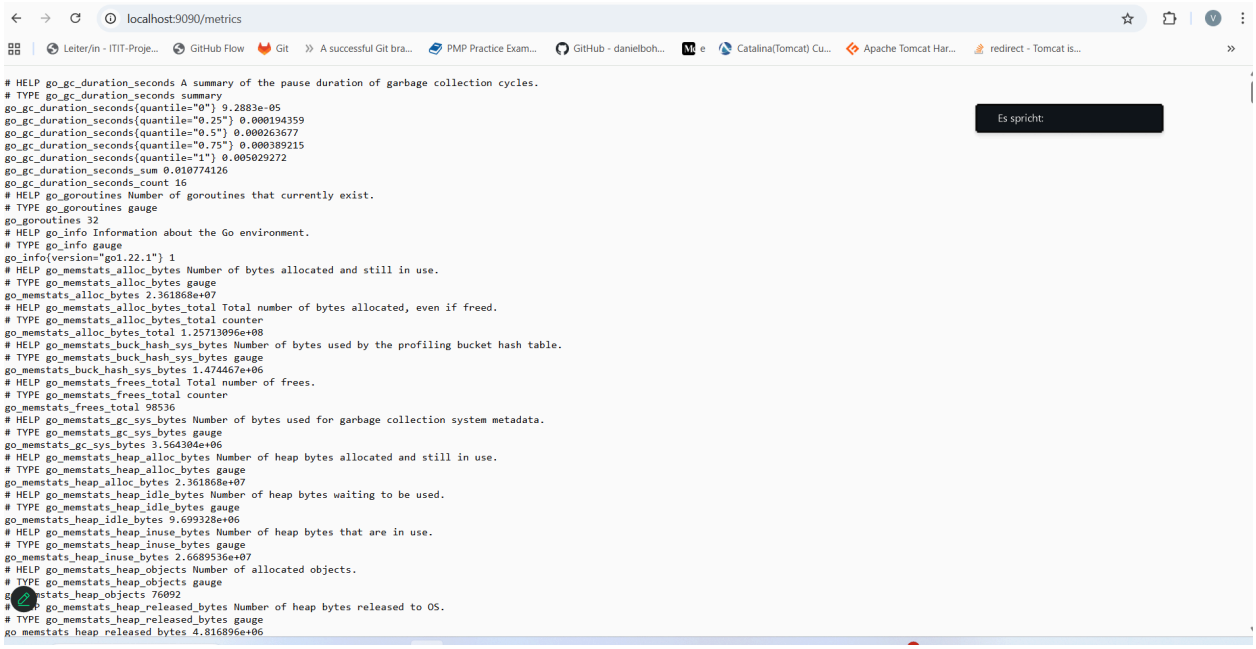
LAB | Metrics from a VM with Prometheus and Grafana

Docker Containers:

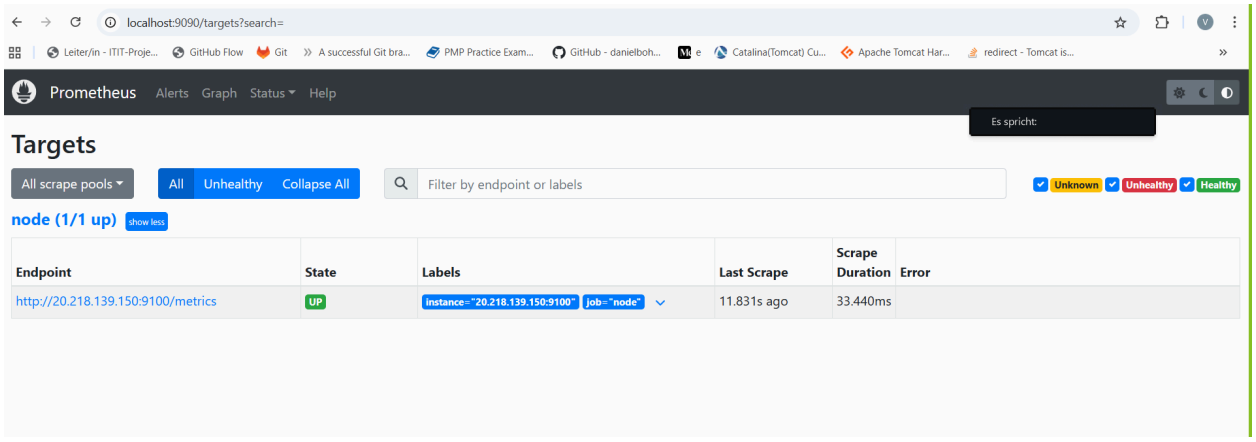
```
vijayagagla@DESKTOP-FP4OP26:~/Metrics-VM$ docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED         STATUS         PORTS                                          NAMES
2b9f904ac0f3   grafana/grafana:latest     "/run.sh"                11 minutes ago  Up 10 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp    grafana
180b9ad7594d   prom/prometheus:v2.51.0    "/bin/prometheus --c…"   11 minutes ago  Up 10 minutes  0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp    prometheus
27e7b20574f6   prom/node-exporter:v1.7.0  "/bin/node_exporter"     11 minutes ago  Up 10 minutes  0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp    node-exporter
```

Prometheus Metrics:

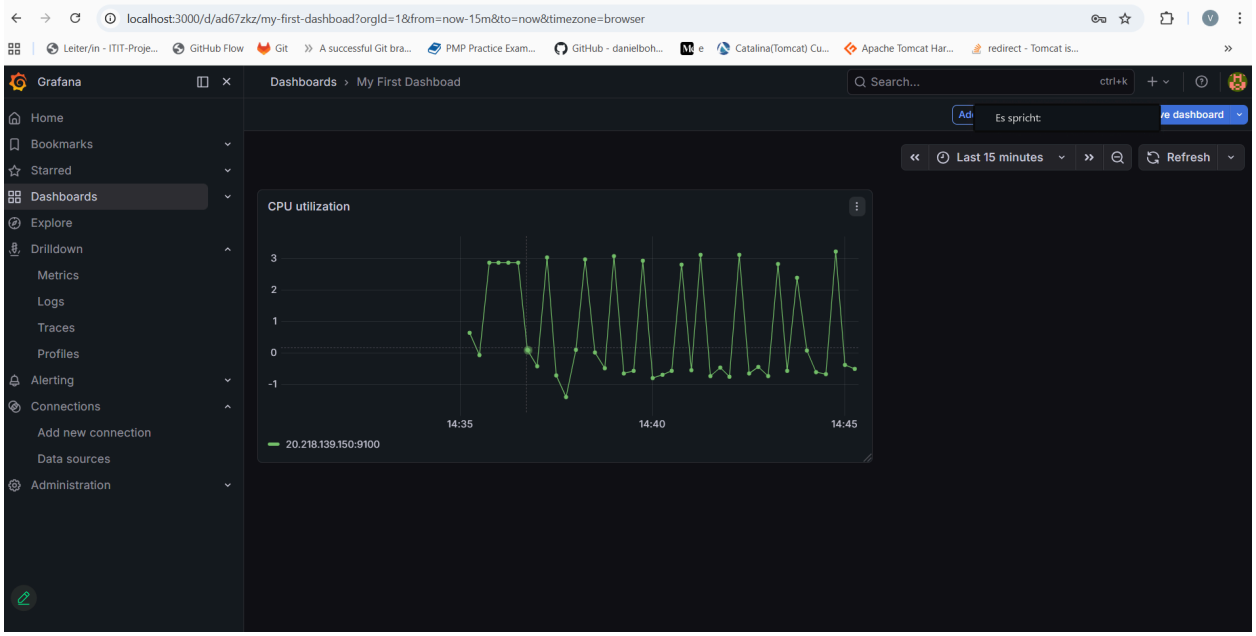



```
# HELP go_gc_duration_seconds A summary of the pause duration of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 9.2883e-05
go_gc_duration_seconds{quantile="0.25"} 0.000194359
go_gc_duration_seconds{quantile="0.5"} 0.000263677
go_gc_duration_seconds{quantile="0.75"} 0.000389215
go_gc_duration_seconds{quantile="1"} 0.005029272
go_gc_duration_seconds_sum 0.010774126
go_gc_duration_seconds_count 16
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 32
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.22.1"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated and still in use.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 2.361868e+07
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated, even if freed.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 1.25713096e+08
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.474467e+06
# HELP go_memstats_frees_total Total number of frees.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 98536
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 3.564304e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and still in use.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 2.361868e+07
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used.
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 9.699328e+06
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use.
# TYPE go_memstats_heap_inuse_bytes gauge
go_memstats_heap_inuse_bytes 2.6689536e+07
# HELP go_memstats_heap_objects Number of allocated objects.
# TYPE go_memstats_heap_objects gauge
go_memstats_heap_objects 76092
# HELP go_memstats_heap_released_bytes Number of heap bytes released to OS.
# TYPE go_memstats_heap_released_bytes gauge
go_memstats_heap_released_bytes 4.816896e+06
```

Prometheus Targets:



Grafana Dashboard for the CPU utilization


localhost:3000/d/ad67zkz/my-first-dashboad?orgId=1&from=now-15m&to=now&timezone=browser
Grafana
Dashboards › My First Dashboad
Search...
ctrl+k
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Metrics
Logs
Traces
Profiles
Alerting
Connections
Add new connection
Data sources
Administration
Es spricht:
Last 15 minutes
Refresh
CPU utilization
3
2
1
0
-1
14:35
14:40
14:45
20.218.139.150:9100